

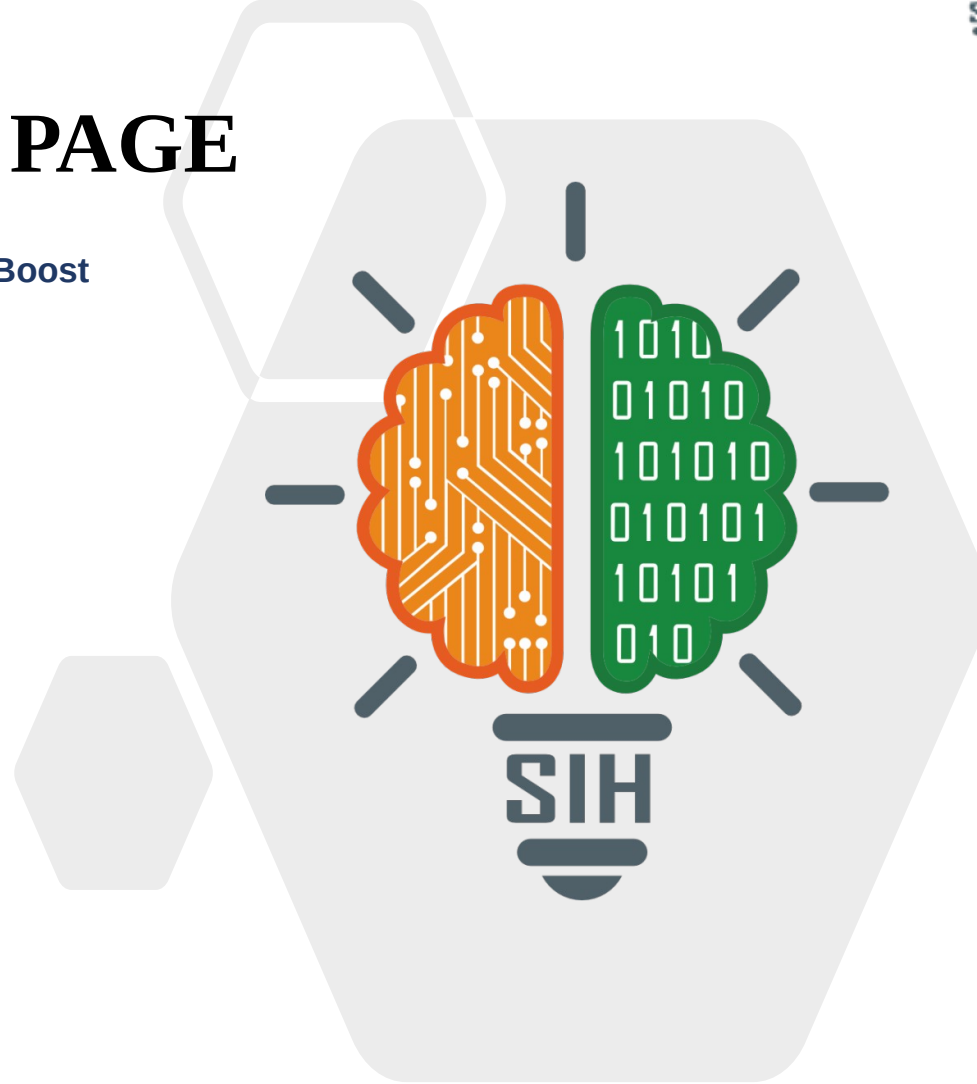
SMART INDIA HACKATHON 2025



TITLE PAGE

Idea: PRAHARI – AI Demand Forecasting & Crowd Rebalancing to Boost Hotels, Homestays and Travel in India

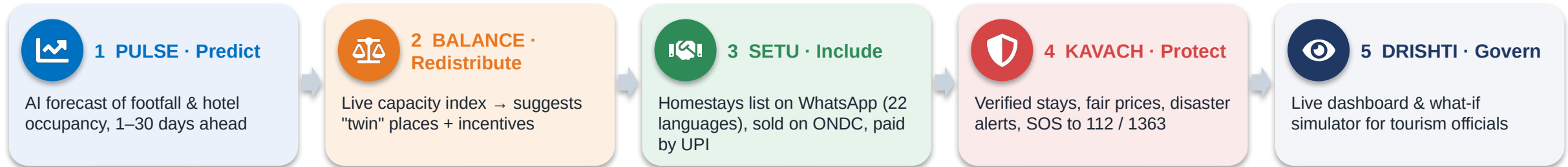
- **Problem Statement ID** – _____
- **Problem Statement Title**- Student Innovation
- **Theme**- Travel & Tourism
- **PS Category**- Software
- **Team ID**- _____
- **Team Name (Registered on portal)** _____



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

❖ Detailed explanation of the proposed solution

PRAHARI (Predictive Routing & AI for Hotels, Attractions, Regions & Itineraries) predicts where and when tourists will go, checks each destination's live carrying capacity, and sends surplus travellers to nearby **"twin" destinations and small hotels with free rooms**.



❖ How it addresses the problem

- **Overcrowded hotspots** → 7–14 days early warning + overflow moved to twins
- **Empty rooms next door** → surplus demand sent to under-used homestays
- **15–25% OTA commission** → single-digit fees via government-backed ONDC
- **Officials act too late** → live dashboard + permit / campaign simulator
- **Tourist safety gaps** → SACHET alerts, SOS, fair-price registry

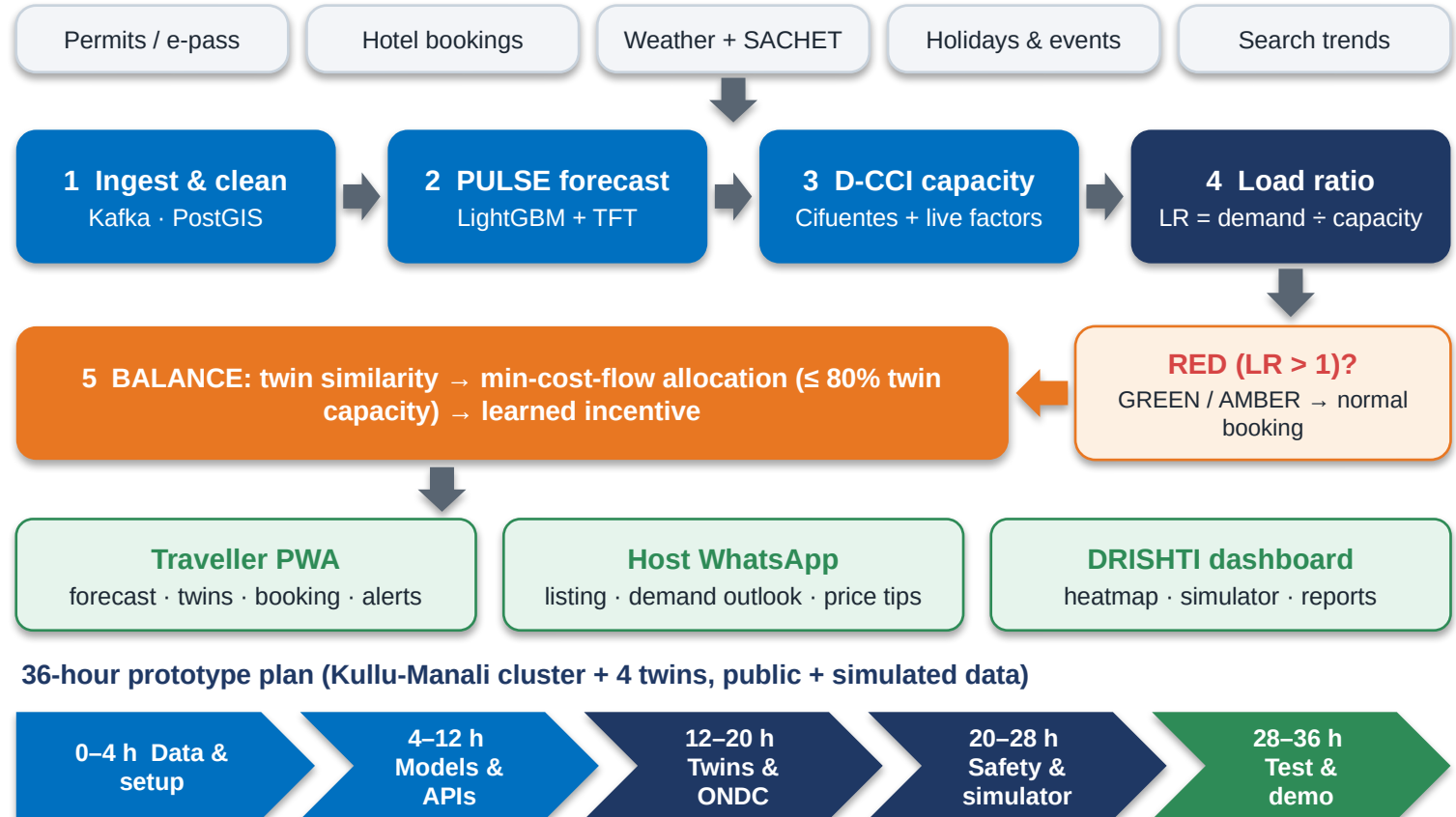
❖ Innovation and uniqueness of the solution

- **First closed loop:** Predict → Measure → Redistribute (others only sell, inform or restrict)
- **Dynamic Carrying Capacity Index:** one-time studies become a daily number
- **No overflow cascade:** twins are never pushed above 80% of their capacity
- **Self-learning incentives:** Thompson sampling picks the nudge that works
- **Built on India's DPI:** ONDC · Bhashini · UPI · NIDHI · SACHET

Technologies to be used (e.g. programming languages, frameworks, hardware)

Layer	Technology
Languages	Python 3.11 · TypeScript · Node.js
Frontend	React PWA (offline) · WhatsApp bot · MapLibre + deck.gl
Backend	FastAPI microservices · Bechn / ONDC adapter
Data	PostgreSQL + PostGIS · TimescaleDB · Kafka · Redis · H3
AI / ML	LightGBM · Temporal Fusion Transformer · OR-Tools · MLflow
Integrations	ONDC · Bhashini · UPI · DigiLocker · IMD · SACHET · 112
Infra	Docker · Kubernetes · MeitY cloud (data in India)
Hardware	None: runs on any smartphone or browser

Methodology and process for implementation (Flow Charts/Images/ working prototype)



D-CCI = $PCC \times \prod(1 - \text{correction factors}) \times \text{management capacity}$; status: GREEN < 0.8 · AMBER 0.8–1 · RED > 1 · BLACK = hazard / closed

Analysis of the feasibility of the idea

Technical ✓

- Proven open-source stack
- Well-established ML models
- Prototype fits in 36 h

Data & APIs ✓

- Public data: Tourism Statistics, ASI, IMD, holidays
- ONDC · Bhashini · UPI sandboxes

Economic ✓

- District pilot $\approx$ ₹30–40 lakh / year
- 5% of one peak weekend retained $\approx$ ₹2 Cr*

Operational ✓

- WhatsApp-first, zero joining fee
- Fits e-pass, NIDHI & portal workflows

36 hours: Working prototype

0–9 months: District pilot:
Kullu-Manali, Nilgiris

9–18 months: 3–5 states as
SaaS tenants

18–36 months: National
tourism DPI

Potential challenges and risks

Strategies for overcoming these challenges

Sparse daily footfall data

Proxy signals (permits, bookings, search) + cold-start transfer learning + state data MoUs

Low digital adoption by small hosts

WhatsApp & voice onboarding in local language, zero fee, camps with SHGs & district offices

Travellers ignore suggestions

Learned incentives, honest delay estimates, twin offers at the e-pass stage

Crowd just shifts to a fragile village

Hard cap at 80% of twin capacity + community consent before listing

Govt. API dependency & privacy

Modular adapters with cached fallbacks; aggregate-only data, DPDP Act 2023 compliant

*Indicative estimate: 1 lakh peak-weekend visitors $\times$ ₹4,000 spend; to be validated in the pilot.

❖ Potential impact on the target audience



Travellers

7–14 days

early warning of crowds

- Less traffic & queues
- Verified stays, fair prices
- Alerts + one-tap SOS



Hotels & homestays

+10–15 pts

off-season occupancy*

- Listed in < 10 minutes
- Single-digit fees vs 15–25%
- Weekly demand & price tips



Tourism departments

~10 min

weekly report (was ~2 days)

- Plan police, buses, sanitation early
- What-if permit simulator
- Track scheme outcomes

❖ Benefits of the solution (social, economic, environmental, etc.)



Social

- Income for rural & women-led homestays
- Digital inclusion in 22 languages
- Less resident resentment



Economic

- Visitors redirected, not turned away
- Hosts save ₹200–400 per ₹2,000 night
- More local spend & GST



Environmental

- Fragile hill & coastal spots stay under capacity
- Less traffic idling & waste peaks
- Longer destination life cycle



Safety & governance

- Disaster alerts in < 5 minutes
- Evidence-based planning
- Supports SDG 8.9, 11, 12.b

*Pilot targets / indicative estimates, to be validated in a district pilot.

Details / Links of the reference and research work



Government data & policy

- Ministry of Tourism: India Tourism Statistics 2024 & Tourism Satellite Account tourism.gov.in
- Swadesh Darshan 2.0, SASCI iconic destinations (PIB releases) pib.gov.in
- Union Budget 2025-26: 50 destinations, MUDRA loans for homestays indiabudget.gov.in
- Madras HC e-pass for Ooty & Kodaikanal (2024); NGT Rohtang vehicle caps
- UNWTO (2018) 'Overtourism?' Understanding & managing urban tourism growth e-unwto.org
- Digital Personal Data Protection Act 2023 meity.gov.in
- ONDC open network (Beckn protocol) ondc.org
- Bhashini language platform (MeitY) bhashini.gov.in
- NDMA SACHET disaster alerts sachet.ndma.gov.in
- NIDHI hospitality database (Ministry of Tourism) nidhi.tourism.gov.in



Research papers

- Song & Li (2008) Tourism demand forecasting review, Tourism Management 29(2) [doi:10.1016/j.tourman.2007.07.016](https://doi.org/10.1016/j.tourman.2007.07.016)
- Bangwayo-Skeete & Skeete (2015) Google data improves arrival forecasts, Tourism Management 46
- Law et al. (2019) Deep learning for tourism demand, Annals of Tourism Research 75
- Lim et al. (2021) Temporal Fusion Transformers, Int. J. of Forecasting [arXiv:1912.09363](https://arxiv.org/abs/1912.09363)
- Wickramasuriya et al. (2019) MinT forecast reconciliation, JASA [doi:10.1080/01621459.2018.1448825](https://doi.org/10.1080/01621459.2018.1448825)
- Cifuentes (1992) Tourism carrying-capacity method, CATIE
- Butler (1980) Tourist Area Life Cycle, Canadian Geographer [doi:10.1111/j.1541-0064.1980.tb00970.x](https://doi.org/10.1111/j.1541-0064.1980.tb00970.x)
- Agrawal & Goyal (2012) Thompson Sampling analysis, COLT

Supporting research: Detailed Business Model Canvas, architecture, competitor analysis & ROI are in our Additional Document (PRAHARI_Additional_Document.pdf).